Excel Data Project: Interactive Salary & KPI Dashboard

1) Introduction

This project presents an interactive Salary Dashboard built in Microsoft Excel to analyze real-world data science and analytics job market trends. The dashboard enables users to explore how job title, country, and job schedule type impact salary levels using dynamic filtering, KPI indicators, and comparative bar charts. The goal is to transform a large, raw dataset into a decision-support tool for job seekers, recruiters, and analysts.

The project emphasizes formula-driven analytics, chart design choices, and usability-focused dashboard design, rather than manual data preparation.

2) Excel Skills Used

The following Excel skills were utilized throughout the analysis and dashboard build:

- Charts – Horizontal bar charts for salary comparisons and job schedule analysis
- Formulas and Functions – MEDIAN, FILTER, UNIQUE, SORT, COUNT, COUNTIFS, XLOOKUP
- Data Validation – Dropdown-driven filtering for interactive user control

3) Data Jobs Dataset

The dataset contains approximately 32,600 rows of real-world job postings from 2023 in the data science and analytics job market.

This dataset supports both salary trend analysis and job market segmentation across regions, roles, work types, and hiring platforms.

4) Dashboard Build

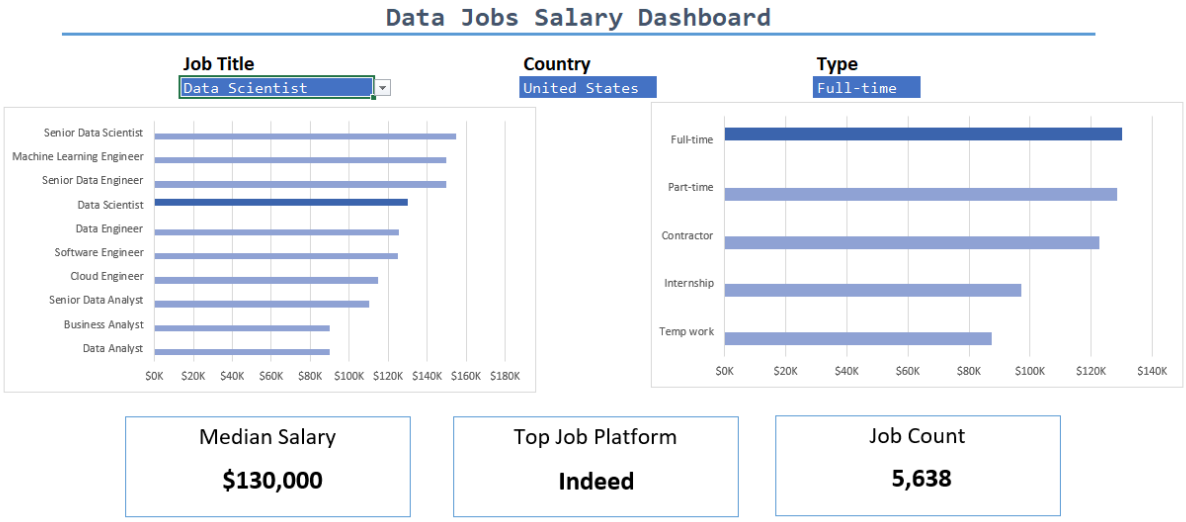
Dashboard Structure

At the top of the dashboard, three interactive dropdowns allow users to filter results by:

- Job Title
- Country
- Job Schedule Type (e.g., Full-Time, Contract, Part-time)

Below the filters, the dashboard displays:

- Two bar charts for salary and job type comparison
- Three KPI cards for high-level performance insights



a) Charts

1. Data Analytics Job Salaries – Bar Chart

Purpose:

Displays median salary by job title, allowing users to compare compensation levels across roles.

Excel Features Used:

- Horizontal Bar Chart
- Salary value formatting
- Dynamic range updates via formulas and dropdown filters

Design Choice:

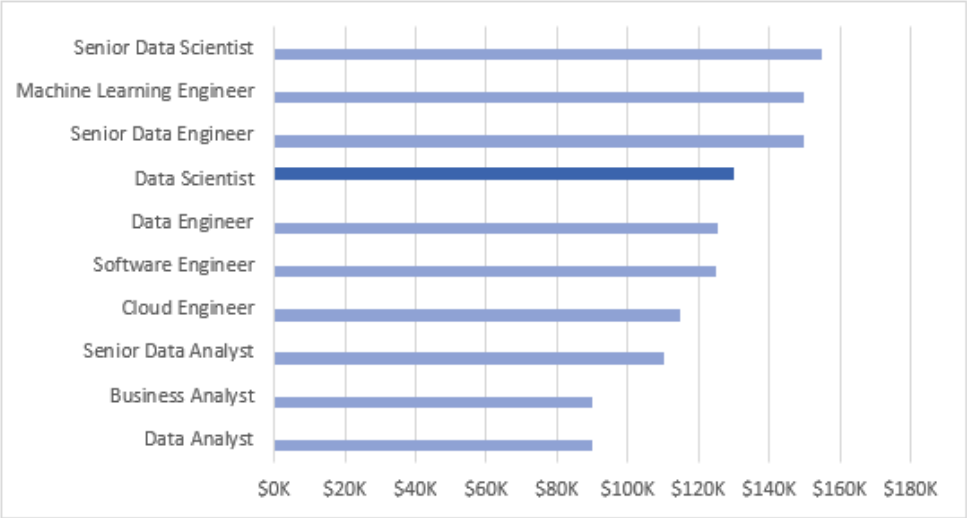
A horizontal layout was selected to improve readability for long job titles and enable faster side-by-side salary comparisons.

Data Organization:

Job titles are sorted by descending median salary using SORT(), ensuring the highest-paying roles are immediately visible.

Insights Gained:

- Senior and Engineering roles consistently appear at the top of salary rankings
- Analyst roles trend lower in median compensation



2. Job Schedule Type vs Median Salary – Bar Chart

Purpose:

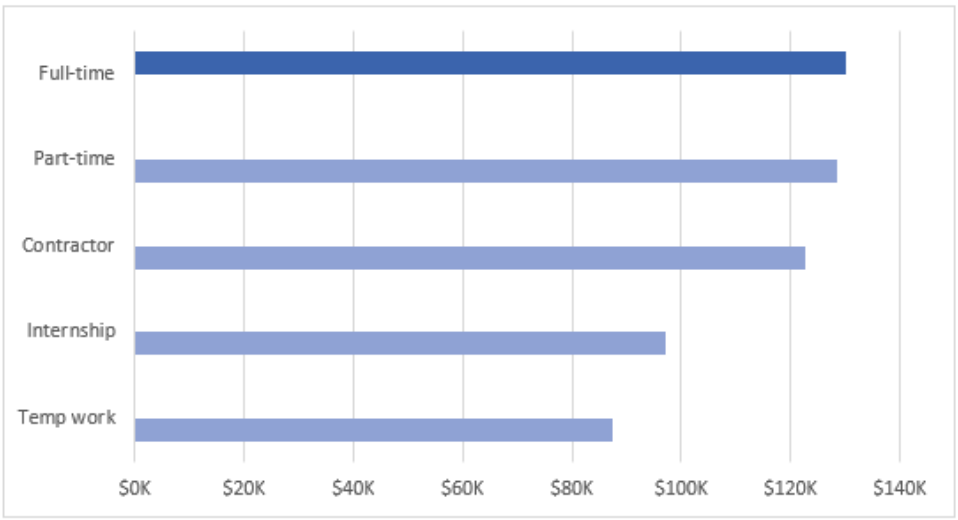
Compares median salary across job schedule types such as Full-Time, Part-time, and Contractor.

Design Choice:

This visualization highlights how work structure impacts compensation, enabling users to identify higher-paying job formats.

Insights Gained:

- Certain flexible or contract-based roles often command lower median pay
- Full-time roles show more consistent and higher salary ranges



b) Formulas and Functions

1. Median Salary by Job Title (Core Analytics Formula)

```
=MEDIAN(
    IF(
        (jobs[job_title_short]=A2)*
        (jobs[job_country]=country)*
        (ISNUMBER(SEARCH(type, jobs[job_schedule_type])))*
        (jobs[salary_year_avg]<>0),
        jobs[salary_year_avg]
    )
)
```

Purpose:
Calculates the median annual salary based on selected job title, country, and job schedule type.

Key Features:

- Multi-Criteria Filtering: Filters by role, region, and work type
- Blank Salary Exclusion: Ensures zero or missing salaries are not included
- Array-Based Analysis: Uses IF() inside MEDIAN() to evaluate multiple conditions simultaneously

Business Value:
Delivers role-specific, location-aware salary insights that reflect real market conditions instead of general averages.

2. Unique Job Schedule Type Extraction

```
=FILTER(J2#,(NOT(ISNUMBER(SEARCH("and",J2#))+ISNUMBER(SEARCH(",",J2#))))*(J2#
<>0))
```

Purpose:
Generates a clean, unique list of job schedule types for dropdown selection.

Key Features:

- Removes combined or invalid entries
- Excludes zero or blank values
- Automatically updates as data changes

Business Value:
Ensures clean, consistent user inputs, improving dashboard reliability and filtering accuracy.

3. Supporting Functions Used

Function	Purpose
COUNT	Measures total job records and platform usage
COUNTIFS	Applies multi-condition job and platform counts
SORT	Orders job titles by salary ranking
UNIQUE	Extracts clean dropdown lists
FILTER	Applies dynamic dataset filtering
XLOOKUP	Links KPI cards to filtered reference tables

c) Data Validation

Filtered Dropdown System

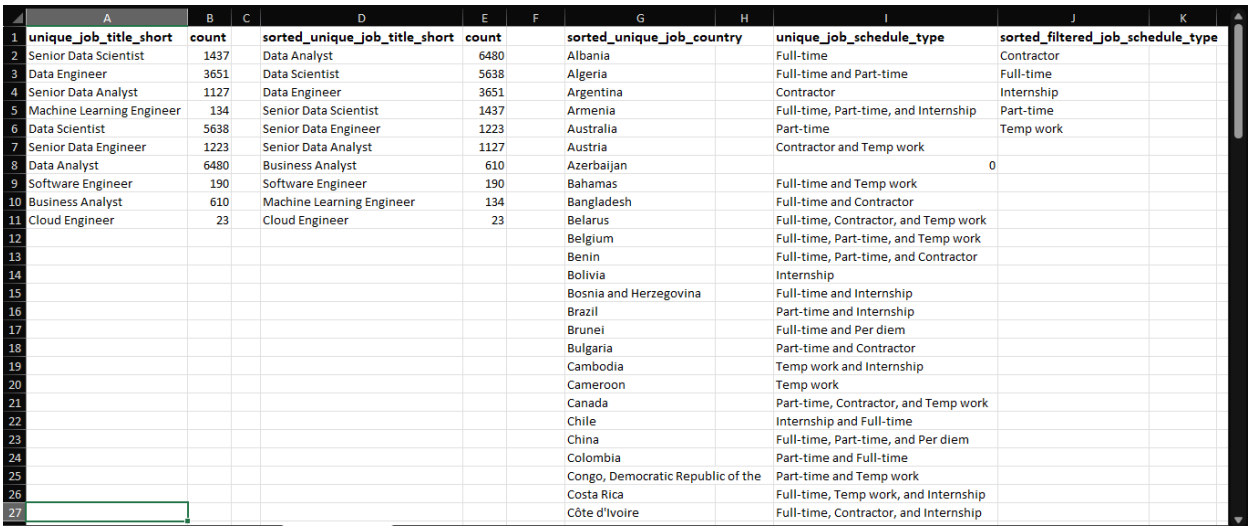
The Data_Validation sheet acts as a centralized source for dropdown values:

- Job Titles
- Countries
- Job Schedule Types
- Platforms

Implementation Benefits:

- Prevents invalid or inconsistent user inputs
- Improves dashboard usability
- Ensures formulas always reference validated values

This structure enables a controlled, professional user experience similar to BI tools such as Power BI or Tableau.



unique_job_title_short	count	sorted_unique_job_title_short	count	sorted_unique_job_country	unique_job_schedule_type	sorted_filtered_job_schedule_type
Senior Data Scientist	1437	Data Analyst	6480	Albania	Full-time	Contractor
Data Engineer	3651	Data Scientist	5638	Algeria	Full-time and Part-time	Full-time
Senior Data Analyst	1127	Data Engineer	3651	Argentina	Contractor	Internship
Machine Learning Engineer	134	Senior Data Scientist	1437	Armenia	Full-time, Part-time, and Internship	Part-time
Data Scientist	5638	Senior Data Engineer	1223	Australia	Part-time	Temp work
Senior Data Engineer	1223	Senior Data Analyst	1127	Austria	Contractor and Temp work	
Data Analyst	6480	Business Analyst	610	Azerbaijan	0	
Software Engineer	190	Software Engineer	190	Bahamas	Full-time and Temp work	
Business Analyst	610	Machine Learning Engineer	134	Bangladesh	Full-time and Contractor	
Cloud Engineer	23	Cloud Engineer	23	Belarus	Full-time, Contractor, and Temp work	
				Belgium	Full-time, Part-time, and Temp work	
				Benin	Full-time, Part-time, and Contractor	
				Bolivia	Internship	
				Bosnia and Herzegovina	Full-time and Internship	
				Brazil	Part-time and Internship	
				Brunei	Full-time and Per diem	
				Bulgaria	Part-time and Contractor	
				Cambodia	Temp work and Internship	
				Cameroon	Temp work	
				Canada	Part-time, Contractor, and Temp work	
				Chile	Internship and Full-time	
				China	Full-time, Part-time, and Per diem	
				Colombia	Part-time and Full-time	
				Congo, Democratic Republic of the	Part-time and Temp work	
				Costa Rica	Full-time, Temp work, and Internship	
				Côte d'Ivoire	Full-time, Contractor, and Internship	

5) KPI Cards

The dashboard includes three dynamic KPI indicators:

1. Median Salary

- Reflects filtered job, country, and schedule type
- Uses XLOOKUP and MEDIAN logic for real-time updates

2. Top Job Platform

- Identifies the platform with the highest job posting volume
- Driven by COUNT and SORT-based ranking

3. Job Count

- Displays the total number of job postings under active filters
- Updates dynamically using COUNTIFS

Median Salary	Top Job Platform	Job Count
$130,000	**Indeed**	**5,638**

6) Conclusion

This Salary Dashboard demonstrates how advanced Excel formulas and thoughtful chart design can transform large job market datasets into an interactive decision-support tool.

By combining dynamic filtering, median-based salary analysis, KPI tracking, and clean data validation, this project delivers reliable insights into:

- Salary differences across job roles
- Regional compensation trends
- The impact of job schedule types on pay
- Platform dominance in job postings

This approach mirrors professional BI workflows while remaining fully Excel-based, making it ideal for career analysis, hiring strategy, and portfolio demonstration of analytical and dashboard design skills.